

Quality Management System Manual

Example LLC

Quality Management System per ISO 9001:2015

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Table of Contents

1.	Quality Policy	3
2.	Quality Objectives	4
3.	Scope of the Quality Management System	5
3.1	Context of the Organisation	5
3.2	Interested Parties	6
3.3	Risks and Opportunities	7
4.	Process Overview	8
4.1	Leadership Processes	9
4.2	Core Processes	10
4.3	Support Processes	11
5.	Roles and Responsibilities	12
6.	Document Control Procedure	13
7.	Internal Audit Procedure	14
8.	Management Review Procedure	15
9.	Continuous Improvement Procedure	16
A.	Appendix A: Sample Inspection Record	17
B.	Appendix B: Supplier Evaluation	18

1. Quality Policy

At Example LLC, every calibration activity is performed with traceability and technical rigor at its core. Our laboratory ensures that all customer equipment—ranging from gauges and torque tools to pressure and temperature instruments, and digital multimeters—is calibrated against reference standards traceable to NIST. Before any equipment leaves our facility or is returned at an on-site job, the corresponding calibration certificate documents precisely what was measured, including method, reference standards, and all relevant calculations or raw readings.

We maintain an uncompromising stance on documentation integrity. Each certificate receives a final review and is only released if we can stand behind every result. If any ambiguity in the calibration data or traceability arises, the job remains open until all issues are resolved. Our operation is structured to withstand scrutiny from customer audits or their end-users' auditors, ensuring consistency no matter the customer's industry—especially those in automotive and aerospace supply chains who require full traceability and fast turnaround.

Meeting promised delivery times is a central aspect of our service model. Delays directly impact our customers' production schedules and can undermine trust. To that end, every calibration request is logged and scheduled upon intake, with responsibilities clearly divided between the office, technical, and oversight staff. Issues such as incorrect values or late shipments escalate immediately for corrective action and root cause review.

We foster organizational quality by integrating feedback from customers, tracking correction rates on certificates, and formally reviewing recurring problems to prevent repetition. Employee training, knowledge transfer, and procedural review are prioritized, ensuring new and experienced technicians adhere to standardized, documented methods. Our quality management system is designed to minimize errors, eliminate overdue reference standards, and ensure that no detail escapes documentation, in line with traceability and documentation expectations established by NIST standards and the requirements of our most demanding customers.

This manual is structured to fulfill the requirements of ISO 9001:2015, chapters 4 to 10.

2. Quality Objectives

QUALITY OBJECTIVES BY END OF 2026

Certificate correction rate	6 → 2 per year
On-time delivery rate	88 % → 95 %
Reference standards past due for recalibration	2 → 0

Our quality objectives are founded on practical performance data and aim to address both customer expectations and internal compliance requirements. The first objective is to reduce the certificate correction rate: in the previous year, six certificates were amended post-issue due to issues such as value transpositions or units errors. Our target is to bring this number down to two or fewer per year by the end of 2026, which we will achieve by emphasizing final reviews and technician training in correctly documenting all measured and calculated results.

A second objective is to improve our on-time delivery rate. In 2025, approximately 88% of calibrations were shipped or delivered by the promised date. Our goal is to reach at least 95% on-time performance by the end of 2026. Every job is logged with a committed due date, and delays are reviewed to ensure that scheduling, workload, or equipment resource issues are identified and resolved quickly.

The third objective addresses the foundation of our traceability claims: all reference standards used in the laboratory must be currently certified and within calibration. In 2025, we had two reference standards overdue for recalibration—unacceptable for our promise of NIST traceability. Our strict target is to maintain zero overdue standards at any time as of Q1 2026. This is monitored through tracking logs, with proactive scheduling for recalibration well before due dates.

Progress towards each objective is monitored regularly in management meetings, with responsible staff assigned to data tracking, the initiation of corrective actions, and procedure improvements as required.

3. Scope of the Quality Management System

The quality management system at Example LLC covers all company activities involving the calibration, documentation, and return of customer measuring and test equipment. This encompasses received equipment at our Greenville, South Carolina laboratory, as well as on-site jobs at customer locations. All calibration disciplines—dimensional, pressure, temperature, and electrical—are included.

Processes for customer inquiry, scheduling, handling incoming and outgoing shipments, performing calibrations with NIST-traceable standards, managing calibration artefacts, issuing certificates, and following up on customer feedback or complaints are part of the system's boundary. The monitoring and recalibration of in-house reference standards, supplier management for external calibration providers, and all supporting documentation processes are explicitly included.

The scope does not extend to external calibration providers' internal processes but includes the interface for delivery, acceptance, and documentation of returned standards and certificates. Warehousing, purely administrative, and sales/marketing activities are excluded except as they directly support calibration fulfillment or documentation accuracy.

3.1 Context of the Organisation

Our business is influenced by the high technical and documentation expectations of the automotive and aerospace supply sectors, with customers frequently demanding calibration certificates and traceability that can withstand full audit scrutiny. Resource and workload planning is shaped by the limited pool of experienced calibration technicians and the dependence on reliable external laboratories for reference standard recalibration. Material and information availability are driven by our reliance on lab management software for records and our carrier's timely delivery network. Staying current against NIST standards and maintaining documented evidence of traceability directly impact our competitive positioning and reputation.

3.2 Interested Parties

Our activities affect and are affected by various stakeholders:

INTERESTED PARTY	KEY REQUIREMENTS	HOW THE COMPANY MEETS THEM
Customers	Precise and traceable calibration, documentation fit for audit, reliable turnaround	NIST-traceable calibrations, strict certificate review, delivery tracking, clear recordkeeping
Suppliers	Accurate scope definition, predictable service demand, timely payment	Scheduled orders for external calibration, clear acceptance protocols, prompt invoice settlement
Employees	Structured procedures, ongoing training, workload clarity, safety	Documented work instructions, technical onboarding, task assignment by discipline, OSHA guidelines
Regulators/ Standard Setters	Traceability to national standards; compliance with lab safety and documentation rules	Calibrations performed to NIST-traceable standards, adherence to OSHA requirements, robust certificate documentation

3.3 Risks and Opportunities

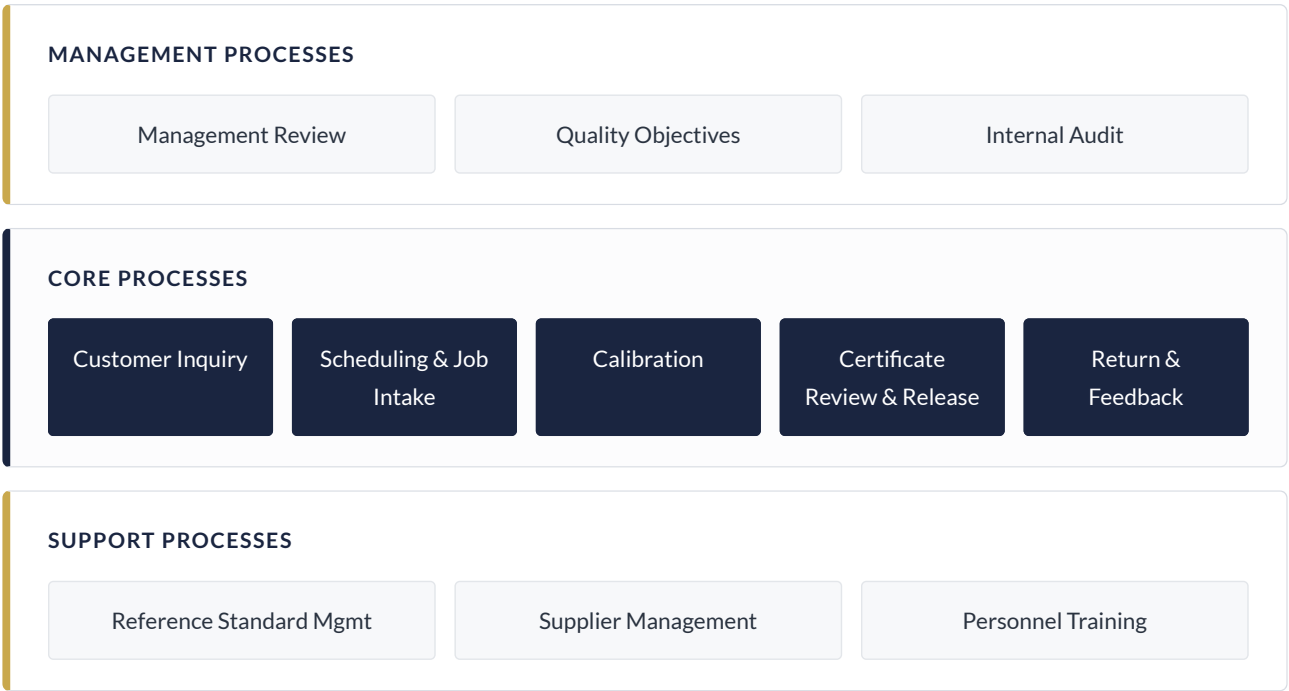
Considering internal and external context, we identify and address the following risks:

RISK	RATING	MEASURE	OWNER
Calibration certificate errors (wrong values/units)	high	Final certificate review & technician training	Marcus Bell
Delays in turnaround during high workload	medium	Resource leveling, cross-training, workload tracking	Dana Whitfield
Reference standards overdue for recalibration	high	Proactive scheduling & escalation for overdue items	Kevin Park
Non-uniform procedures among technicians	medium	Procedure documentation, routine audits, onboarding	Marcus Bell
Supplier failure to return standards on time	medium	Multiple provider relationships, order follow-up	Dana Whitfield

Opportunities include leveraging technician cross-training to flex more efficiently during demand fluctuations and formalizing procedural documentation to make onboarding and quality checks more robust as the team grows.

4. Process Overview

The quality management system at Example LLC is built on integrated processes connecting leadership decisions, core technical execution, and support activities. Leadership processes define direction and control; core processes deliver service; support processes ensure equipment, data, and knowledge flow consistently.



Customer requirements flow through core processes, supported by management and support activities.

4.1 Leadership Processes

Leadership at Example LLC, overseen by Dana Whitfield (Owner), sets goals and monitors performance through recurring management meetings and annual system reviews. At the beginning of each year, Dana, together with Marcus Bell (Lab Manager), reviews quality objective data from the previous twelve months—focusing on certificate correction rates, on-time delivery percentages, and reference standard calibration status. This review is triggered by the preparation of the annual quality summary, which is compiled from lab management software reports and technician logs.

During these sessions, performance against defined targets is assessed. When deviation from targets, such as an increased correction rate or a missed delivery metric, is identified, Dana initiates a review with relevant team members to determine whether the cause is rooted in resource planning, procedural shortcomings, or training requirements. Decisions on corrective and preventive actions are reached jointly by Dana, Marcus, and Theresa Nguyen when customer communication or process adjustment is needed.

Leadership is also responsible for resource decisions. When workload imbalances or technician shortages are detected—usually flagged by Marcus or by tracking in lab scheduling software—Dana steps in to adjust resource distribution or to approve overtime, outside training, or temporary staff as needed. Supplier risks are elevated for decision when recalibration delays threaten NIST traceability, often based on a report by Kevin Park. These cases are discussed directly between Dana and Marcus, who agree on escalation paths, such as changing accredited suppliers or communicating risk to customers.

Any recurring or high-severity deviations are tracked in a centralized log, with responsibility for follow-up assigned during meetings. Completed actions and progress against each objective are reviewed, and unresolved issues are carried forward with clear next steps. Leadership documents every decision and includes these reviews as part of the annual management review, ensuring issues do not fall through the cracks.

4.2 Core Processes

Core processes start with the customer's calibration request, which may be received by phone, email, or through an online form. Upon receiving a request, Theresa Nguyen logs the job into the lab management software and confirms with the customer the scope, type of instruments, and required turnaround time. When equipment is shipped to the lab, Theresa checks all incoming items, noting serial numbers and any shipping damage. For on-site calibrations, a visit is scheduled with the customer, and a lead technician is assigned in advance.

Marcus Bell reviews the incoming workload and assigns each job to a technician matched with the required measurement discipline (dimensional, pressure, temperature, or electrical). The technician checks that the appropriate NIST-traceable reference standards are available and within calibration period before proceeding with the job. Throughout the calibration, technicians record all measured values directly into the lab management software and retain physical bench notes as backup.

After calibration, the instrument is marked with a calibration label stating the date, standard used, and next recalibration due. The draft certificate is generated by the software and is reviewed—either by Dana or, in routine cases, as a spot-check. Any discrepancies found in the certificate or its traceability path require the technician to revisit measurements or correct data entries before the job is closed.

Upon approval, Theresa prepares outbound shipping, including the signed calibration certificate and updated labels, and ensures prompt return to the customer's designated contact. For on-site calibrations, the technician delivers the certificate and equipment directly before departure, confirming acceptance with the customer. All job records and certificates are stored for traceability and later audit.

When a customer disputes or requests a correction to a certificate—such as noting a wrong value or missing measurement—Theresa captures the complaint in the tracking system and notifies Dana and Marcus. The original technician is asked to retrace their steps, consult bench notes, and clarify the error. Corrections are issued with an explanatory note, and patterns of error are escalated to Marcus for root cause review and, if required, written corrective actions.

4.3 Support Processes

Support processes provide the foundation for accuracy, safety, and reliability across calibration workflows. The management of reference standards is led by Kevin Park, who maintains an up-to-date tracking log. Sixty days before any reference standard is due for recalibration, the tracking system flags the standard, and Kevin plans shipment to an accredited external calibration provider. Upon return, Kevin logs the updated calibration certificate, checks conformance of the new values, and links the standard's status to calibration scheduling. When a standard is found out of tolerance, Kevin immediately escalates to Marcus and Dana for joint review, potentially triggering an impact analysis of work performed with the affected reference.

Document control is essential, especially given strict audit requirements from customers. All calibration and procedural documents are version-controlled in the central document library, with Marcus authorizing any updates or revisions. When changes affect standard operating procedures, Marcus leads a review session with all technicians to explain the changes and check understanding. Only current documents are accessible in the lab or on-site via secure access, reducing the risk of outdated instructions.

Purchasing and monitoring of external calibration service providers fall to Dana and Kevin. Orders are placed with suppliers according to the annual schedule, with every shipment and return tracked. Delays or quality issues from suppliers are recorded, triggering a review if a provider fails to meet deadlines or documentation expectations. Regular supplier evaluation ensures that only accredited partners who meet audit and turnaround requirements are retained.

Personnel support is addressed through ongoing training, onboarding, and supervision. New or trainee technicians are assigned to shadow experienced staff, with Marcus overseeing their integration and confirming comprehension via routine checks on bench work and documentation. Knowledge is transferred through structured onboarding checklists and regular briefings, making organizational know-how not dependent on individual experience. Periodic workflow checks are performed to ensure all staff have mastered core and support procedures.

The above ensures that document revisions, supplier reliability, standard maintenance, and staff competency directly support technical consistency and service quality throughout the company.

5. Roles and Responsibilities

Dana Whitfield, as Owner, leads the company with responsibility for overall strategic direction, resource management, and compliance with customer and regulatory requirements. Dana directly oversees the quality management system, initiates annual management reviews, and chairs meetings where corrective actions or resource adjustments are decided. Dana also makes final decisions regarding supplier changes, escalation of customer issues, and significant procedural updates, coordinating closely with Marcus Bell and the wider team.

Marcus Bell, Lab Manager and designated Quality Manager, acts as the operational lead on technical quality, overseeing the daily calibration work and verifying adherence to procedures. Marcus is responsible for matching technicians to calibration jobs based on discipline and skill, reviewing or spot-checking calibration certificates prior to release, and leading root cause investigations for any customer complaints. Where corrective or preventive actions are required—such as for recurring certificate errors or process nonconformities—Marcus leads documentation, execution, and closure. Marcus also supervises the onboarding and ongoing training of new or less-experienced technicians, ensuring process knowledge is consistently spread across the team and not left to individual interpretation.

Theresa Nguyen provides administrative and logistical support, handling customer inquiries, job intake, invoicing, and the entire cycle of inbound and outbound shipment of equipment. Theresa also maintains the job tracking system, logs customer complaints, and ensures that certificates and job records are properly routed and filed after closure. When customer feedback or complaints are received, Theresa initiates the notification process to Dana and Marcus.

Kevin Park maintains oversight of all in-house reference standards, including their calibration status, due dates, and shipment to external accredited providers. Kevin ensures that no standard falls overdue and manages the corresponding documentation. Any discrepancies or delays in the standard return process are escalated by Kevin directly to Marcus or Dana for decision.

Technicians are responsible for executing calibrations strictly according to documented procedures, recording all bench measurements and results in lab software and physical notes. Each technician checks their assigned reference standards before use, ensuring all are traceable and currently certified. Trainees operate under Marcus's supervision until workflow checks are passed.

Organizational knowledge is transferred formally: every new technician or trainee follows a structured onboarding checklist, and experienced staff anchor hands-on checks directly in the workflow, so procedural knowledge is never solely dependent on individual memory or informal habits.

6. Document Control Procedure

Document control at Example LLC ensures that all calibration procedures, technical instructions, and quality records remain current, accessible, and traceable. Any creation or revision of documents—ranging from calibration procedures to corrective action reports—is initiated by Marcus Bell. Drafts are reviewed and, if changes affect core procedures, discussed with the technicians in a briefing session.

The approved version is uploaded to the document library, with obsolete versions removed from operational folders. The document library is managed electronically with version tracking; only current versions are accessible from lab terminals or via remote access for on-site work. If a technician identifies a gap, ambiguity, or outdated step during operations, the issue is reported to Marcus, who updates procedures accordingly. A change log records all updates, who made them, and the reason for change.

Calibration certificates, bench notes, and job records are stored both digitally and, where required by audit policy, as physical archives, ensuring traceability for a minimum retention period of five years. Any requested copies for audits are generated from the secured archive.

Document distribution to staff is managed by notifications and regular update briefings, reducing chances of outdated instructions circulating informally. Access controls prevent unauthorized edits or deletions.

7. Internal Audit Procedure

Internal audits are scheduled annually and may be supplemented in response to repeated errors or significant process changes. Dana Whitfield appoints an auditor (different from the usual workflow for independence), who reviews a sample of calibration jobs, reference standard files, document control logs, and certificate records.

The audit is triggered ahead of the annual management review and uses checklists aligned with ISO 9001:2015 requirements and NIST documentation needs. The auditor verifies that calibrations were executed per procedure, reference standards were within date, documents were controlled, and customer feedback followed through. Any finding is classified by number and severity, recorded per audit, and summarized in a register that is compared across audit years to identify persistent issues or improvements.

Nonconformities are documented alongside required corrective actions, with responsible roles assigned a target completion date. The outcome of the audit is discussed in management review and drives updates to objectives, procedures, and training.

8. Management Review Procedure

Management reviews occur at least annually and encompass all aspects of the quality management system. Dana Whitfield schedules the review and compiles performance data on quality objectives, audit findings, corrective actions, workload capacity, and supplier reliability. Marcus Bell provides analysis of process performance and technician discipline, while Kevin Park updates on reference standards and supplier adherence.

For customer satisfaction, feedback is consolidated from customer commissioning surveys, complaint logs, and handover discussions. Patterns of customer requests or complaints are tracked, and their impact on quality objectives is considered. This data is used as the main quantitative measure of customer satisfaction.

The meeting addresses resources (people, skills, equipment), changes to procedures, supplier changes, and regulatory or customer requirements. For every open corrective or preventive action, progress is reviewed, and next steps are clearly recorded. If objectives are not met, reasons are analyzed in depth, and new actions, targets, or resource decisions are documented.

Results from the management review—actions taken, objectives updated, resource commitments—are formally disseminated to relevant staff, and any significant process changes are scheduled for retraining and communication.

9. Continuous Improvement Procedure

Continuous improvement is embedded in daily operations and formally reviewed through the audit and management review cycles. When incidents, complaints, or nonconformities arise, immediate action is taken to correct the issue, but the root cause is always examined during follow-up.

Improvement opportunities are identified from trends in correction rates on certificates, missed deliveries, or overdue reference standards, as well as through periodic hands-on workflow checks and staff feedback sessions. Marcus Bell compiles these insights, and Dana reviews in monthly meetings.

If improvement actions require procedural change or resource investment, Dana evaluates feasibility, and Marcus implements pilot changes (e.g., updating a procedure or rolling out a new training module). The effect is tracked over a defined period—such as a reduction in recurring errors or a boost in on-time delivery—and incorporated into the annual objectives review.

Any change made for improvement is officially logged, and all staff are briefed once the update is finalized, ensuring organization-wide learning.

Appendix A: Sample Inspection Record

Final instrument inspection steps are documented as follows:

INSPECTION STEP	RESULT	REMARKS
Equipment received and identified	Pass	Serial number matches job log
Reference standards verified	Pass	Within calibration date
Measurement procedure followed	Pass	Check against current method
All data recorded in software	Pass	Bench notes attached
Certificate reviewed and printed	Pass	Ready for customer delivery

Appendix B: Supplier Evaluation

Main external calibration and logistics suppliers are evaluated on the criteria below:

SUPPLIER	QUALITY	DELIVERY RELIABILITY	STATUS
Accredited calibration provider A	Consistent	On-time last 12 months	Approved
Accredited calibration provider B	Occasional late certs	Two late returns, Q2 2025	Under review
Shipping carrier	Reliable	Minor delays in Q4 peak	Approved